

# Coloring Cubes

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A cube has six faces. If you have six different colours and use one colour on each face, how many different cubes can you make? To answer this, we first have to decide what “different” means, and to help with that we will construct an origami cube.

## *Activity 1:*

Work in pairs. Each person makes 3 Sonobe units. Put the six units together to build an origami cube. As it's paper, we can mark/write on the faces of the cube.

## *Activity 2:*

One student closes their eyes while the other either rotates the cube, or leaves it as it was. The first student opens their eyes and has to tell whether the cube has been rotated or not.

We then explore the different rotations of a cube and discover that there are 24 ways to rotate it without changing its overall structure. We informally explore what happens when two rotations are performed one after another.

At this point we will return to the original question. We can expect two ways of answering the question. One, the 6 faces can be painted in  $6!$  ways. Each cube is counted 24 times, so the number of different ways  $= 6!/24$ . Two, fix a face, pick the opposite face in 5 ways, the remaining 4 faces form a circular permutation in 6 ways, giving a total of  $5 \times 6$  ways. This bit in my experience takes the most amount of time. We can do problem 2 and more if we have time, if not, the rest will be homework.

## *Problem 2:*

Instead of 6 colors, what if we had only 2 colors and had to color 3 faces in one color and the other 3 in the other color. How many different cubes can be made like that?

## *Problem 2:*

What if we had only 3 colors and had to color 2 faces in each color. How many different cubes can be painted like that?

## *Activity 3:*

If time permits, create triangle units to make a tetrahedron. Repeat problem 1, 2, 3 and Activity 2.

## *Homework:*

Use the triangle units to create an icosahedron. Repeat problem 1, 2, 3 and Activity 2.